Flashcards — CampusX 100 Days of ML | Day 13–20

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Day 13 **Q1**

What is the universal sklearn workflow? Name all 5 steps.

ANSWER

1. Load and inspect data (head, info, shape). 2. Separate features X and target y. 3. Split into train/test using train_test_split. 4. Instantiate model → fit on train → predict on test. 5. Evaluate using a metric (accuracy_score, etc.).

Day 13 **Q2**

What do test_size and random_state mean in train_test_split?

ANSWER

test_size: fraction of data held out for testing. 0.2 = 20% test, 80% train. random_state: seeds the randomizer so you get the same split every run (reproducibility). Without it, the split changes each run and results are not comparable.

Day 13 **Q3**

What are the 3 core methods every sklearn model has?

ANSWER

.fit(X_train, y_train) — trains the model. .predict(X_test) — generates predictions. .score(X_test, y_test) — evaluates performance. These 3 methods work identically for ALL sklearn algorithms — only the class name changes.

Day 14 **Q4**

What are the 4 main types of data in ML?

ANSWER

Numerical Continuous: any value in a range (height, salary, temperature). Numerical Discrete: countable integers only (clicks, students, age). Categorical Ordinal: categories with a natural order (rating 1-5, size S/M/L/XL). Categorical Nominal: categories with NO order (city, gender, blood group, color).

Day 14 **Q5**

What is the difference between structured and unstructured data?

ANSWER

Structured: tabular format — rows and columns. Stored in CSV or SQL. Traditional ML algorithms work directly on it. Makes up ~20% of real-world data. Unstructured: no fixed format — images, audio, video, raw text. Makes up ~80% of real data. Needs DL (CNN, Transformer) or feature extraction first.

Day 14 **Q6**

What is the difference between an ML Metric and a Business Metric?

ANSWER

ML Metric: what the model optimizes — accuracy, F1, RMSE, AUC. Business Metric: what the company actually cares about — revenue, customer retention, cost. They are NOT the same. A model with 99% accuracy can still fail on the business goal. Always define both before starting any project.

Day 15 **Q7**

Name 5 important pd.read_csv() parameters and what each does.

ANSWER

sep: column separator character (default comma, use '\t' for TSV). index_col: which column to use as the row index. usecols: load only specific columns — saves memory. nrows: read only the first N rows. na_values: list of strings to treat as NaN (e.g., ['?', 'NA', '--']).

Day 15 **Q8**

How do you handle a CSV file too large to fit in RAM?

ANSWER

Use the chunksize parameter in pd.read_csv(). It returns an iterator — each iteration gives you a DataFrame chunk of that many rows. You process each chunk and discard it before loading the next. This lets you work with terabyte-scale files on a normal machine.

Day 16 **Q9**

What is JSON and how is it related to Python data structures?

ANSWER

JSON (JavaScript Object Notation) is a text format for structured data using key-value pairs. It maps directly to Python: JSON object → Python dict, JSON array → Python list. It's the universal format for APIs and web services. pd.read_json() loads it directly into a DataFrame. For nested JSON, use pd.json_normalize() to flatten it.

Day 16 **Q10**

What is the SQLite + pandas workflow for reading data from a database?

ANSWER

1. Connect: sqlite3.connect('database.db'). 2. Query: pd.read_sql_query('SELECT * FROM table', conn) → returns a DataFrame. 3. Close connection. To write a DataFrame back to SQL: df.to_sql('table_name', conn, if_exists='replace'). SQLite needs no server — it's a single file database, perfect for local ML workflows.

Day 16 **Q11**

When would you use CSV vs JSON vs SQL for ML data?

ANSWER

CSV: flat tabular data, easy to share, works with pandas directly. JSON: API responses, nested or hierarchical data, web service output. SQL: data with relationships between tables, need to query subsets efficiently, production databases where data is updated continuously.

Day 17 **Q12**

What is a REST API and what HTTP status codes must you know?

ANSWER

A REST API is a web interface for fetching/sending data over HTTP. You send a GET request to a URL endpoint and get a JSON response. Status codes: 200 = OK (success). 400 = Bad Request. 401 = Unauthorized (wrong/missing API key). 404 = Not Found. 500 = Server Error. Always check status_code == 200 before parsing the response.

Day 17 **Q13**

What is an API key and how do you use it safely?

ANSWER

An API key is a unique token that identifies you to the API and controls access. It goes in the request params or headers. Never hardcode it in your script — use environment variables or a .env file. Rate limiting: most APIs limit requests per minute — add time.sleep() between calls.

Day 18 **Q14**

What is web scraping and what are the two main Python libraries for it?

ANSWER

Web scraping: automatically extracting data from websites by parsing their HTML. Used when no API exists and data is on a public website. Libraries: requests (downloads the raw HTML of the page) + BeautifulSoup (parses the HTML to find and extract specific elements). Always check robots.txt first to confirm scraping is allowed.

Day 18 **Q15**

What are the 4 key BeautifulSoup methods to know?

ANSWER

soup.find(tag) → returns the first matching tag. soup.find_all(tag) → returns a list of all matching tags. tag.text → extracts the visible text content from a tag. tag.get('href') → extracts a specific attribute value (e.g., a link URL). Add class_='classname' or id='id' to filter by CSS class or HTML id.

Day 18 **Q16**

When would you use pd.read_html() instead of BeautifulSoup?

ANSWER

When the website has proper HTML tags. pd.read_html(url) reads ALL tables on the page and returns a list of DataFrames. Much faster than writing custom scraping code. Works well on Wikipedia, financial data sites, sports stats pages. Use BeautifulSoup when data is NOT in a table (custom HTML structure).

Day 19 **Q17**

What is the difference between mean, median, and mode? When to use each?

ANSWER

Mean (average): sum / count. Sensitive to outliers. Use when data is symmetric with no extreme values. Median: middle value when sorted. NOT affected by outliers. Use for skewed data like salary, house prices. Mode: most frequent value. Use for categorical data or discrete data.

Day 19 **Q18**

What does skewness tell you about a distribution?

ANSWER

Skewness measures how asymmetric the distribution is. Skewness ≈ 0 → symmetric (normal). Positive skew (>0) → long right tail → Mean > Median > Mode. Common examples: salary, house prices (few very high values pull tail right). Negative skew (<0) → long left tail → Mean < Median < Mode.

Day 19 **Q19**

What is IQR and why is it a better spread measure than Range?

ANSWER

IQR = Q3 - Q1 (the middle 50% of data). Range = Max - Min. A single outlier completely changes the Range. IQR ignores the top and bottom 25% — so one extreme value has zero effect. This makes IQR a robust measure of spread. Outlier rule: any point beyond Q1 - 1.5×IQR or Q3 + 1.5×IQR is an outlier.

Day 20 **Q20**

What is Univariate Analysis and what are the best plots for numerical vs categorical features?

ANSWER

Univariate: analysing ONE variable at a time. Numerical: Histogram (distribution shape and frequency), Box Plot (outliers, IQR, median), KDE Plot (smooth distribution curve). Categorical: Count Plot / Bar Chart (frequency per category), value_counts(normalize=True) for proportions.

Day 20 **Q21**

What does a Box Plot show? Name all 5 components.

ANSWER

Q1 (25th percentile): bottom edge of the box. Q2 / Median (50th percentile): line inside the box. Q3 (75th percentile): top edge of the box. Whiskers: extend from Q1 - 1.5×IQR (lower) to Q3 + 1.5×IQR (upper). Outliers: individual dots plotted beyond the whiskers.

Day 20 **Q22**

What 5 things should you look for during Univariate EDA?

ANSWER

1. Distribution shape — normal, right/left skewed, or bimodal (two peaks). 2. Outliers — visible beyond box plot whiskers or far tails in histogram. 3. Impossible values — negative age, zero salary for an employed person. 4. Missing values pattern — are they random or concentrated in certain ranges? 5. Class imbalance (for target column) — 95% one class means accuracy is misleading.

Rule: Cover the answer, read the question, say the answer out loud, then check. Mark 'known' only if you got it right without peeking.